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Brain Activity During Meditation in First-Time Meditators

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Abstract

The electrophysiological correlates of meditation states in both short and long-term meditators have been increasingly documented; however, little is known about the brain activity associated with first-time meditation experiences. The goal of this study is to investigate the electrophysiological correlates of a single guided mindfulness meditation session in subjects with no previous meditation experience. We analyzed electroencephalogram (EEG) changes in signal power, hemispheric asymmetry, and information flow between EEG channels, in 16 healthy subjects who were new to meditation practice. Our results show that information flow decreases in the theta (4–8 Hz) and alpha ranges (8–13 Hz) during mindfulness meditation exercise as compared to control: a passive listening condition. These changes are accompanied by a general trend in the decrease of alpha power over the whole scalp. One possible interpretation of these results is that there is an increased level of alertness/vigilance associated with the meditation task rather than reaching the target state. Our study expands on the existing body of knowledge concerning neural oscillations during breathing meditation practice by showing that in participants with no previous meditation training, EEG correlates are different from the electrophysiological signatures of mindfulness meditation found in studies of more advanced practitioners.

Keywords: electroencephalography, mindfulness, meditation, EEG asymmetry, Directed Transfer Function

Introduction

A number of studies revealed measurable EEG changes associated with various forms of mental training such as hypnosis (Tebecis *et al.*, 1975; Sabourin *et al.*, 1990; De Pascalis, 1993), autogenic training (Jacobs and Lubar, 1989) and biofeedback based relaxation techniques (Jacobs and Friedman, 2004). In the last two decades, there has been growing interest in neurophysiological processes occurring during mental training associated with mindfulness meditation. Mindfulness is defined as “the awareness that emerges through paying attention on purpose, in the present moment, and non-judgmentally” (Kabat-Zinn, 1982). Mindfulness training has proven to be an efficacious therapeutic tool for a variety of clinical and nonclinical health problems, and a booster for well-being (see Goyal *et al.*, 2014 for meta-analysis).

Typically in the studies of the electrophysiological correlates of mindfulness meditation the experimental paradigm involves a group of individuals who follow mindfulness training for 6–8 weeks and a wait-list control group (Davidson *et al.*, 2003). The measurements are performed before and after the intervention and sometimes include also a follow-up period. Another class of studies compares data from a group of meditators with data from a control group of non-meditators. Both types of studies convincingly showed that meditative state corresponds to a different state of consciousness characterized by oscillatory EEG correlates. Although the specific state-related changes may depend on the type of meditation (Travis and Shear, 2010; Lee *et al.*, 2018), there are some common effects reported in a large number of studies. They are mainly related to increase in the alpha power (Delmonte, 1984; Aftanas and Golocheikine, 2001) and increase of theta activity

especially in the frontal and midline areas (Aftanas and Golocheikine, 2001; Kubota *et al.*, 2001; Baijal and Srinivasan, 2010; DeLosAngeles *et al.* 2016). Additionally during focused attention meditation increase of gamma power has been often observed (Lutz *et al.*, 2004; Braboszcz *et al.*, 2017). Furthermore, after mindfulness training, a relative left-sided versus a right-sided decrease of alpha power in frontal and central regions was observed (Davidson *et al.*, 2003). It is repeatedly reported that measurable EEG effects increase with the duration of meditation practice (Aftanas and Golocheikine, 2001; Lutz *et al.* 2004). However, it is less clear whether a meditative state may be achieved without prior meditative training and if so, how it would be manifested in neural oscillations measured with EEG. Hence, the goal of this study was to investigate EEG correlates of a meditative state achieved by first-time meditators during guided mindful breathing exercise in comparison to the control condition.

Accordingly, the current study was carried out in novice meditators without any prior meditation experience. In such a group of subjects, we investigated various characteristics of EEG activity recorded during active listening of the guided breathing meditation instructions and during passive listening of a literature excerpt. We focused on changes in EEG power, hemispheric asymmetry, and information flow (directed effective connectivity) between EEG channels for theta (4–8 Hz), alpha (8–13 Hz), and gamma (30–45 Hz) bands. To determine the directions of EEG activity flow, the directed transfer function (DTF) method was applied (Kaminski and Blinowska, 1991). The “direction of EEG activity flow” means that knowledge about the activity at the source EEG channel helps to predict the activity at the target channel. This method reveals information propagation through short and long-range cortico-cortical connections while neglecting passive propagation due to volume conduction, as instantaneous transmissions (zero delay) between the channels are ignored. In many studies, mindfulness meditation was related to changes in activity in the frontal brain areas (reviewed in Tang *et al.*, 2015). Accordingly, in the analysis of information flow between the EEG channels, we grouped EEG electrodes in three brain regions: frontal, central, and parietal-occipital, and then analyzed the activity flow between these different regions.

Our working hypothesis was that novice meditators would be able to follow guided meditation instructions and reach meditation-related state, which is distinct from passive listening and is characterized by different electrophysiological correlates. Specifically, we hypothesized that during the guided meditation task the participants would exhibit qualitatively similar EEG changes, although possibly to a smaller degree, as those described in other studies of EEG during mindfulness meditation. However, we could not formulate firm expectations, as previous work evaluated mostly EEG patterns in more experienced meditators, typically measured following 8 weeks of mindfulness training.

Methods

Participants

Sixteen right-handed healthy persons (6 male and 10 female), aged 22–28 years (23.31 ± 2.15), took part in the experiment. All subjects were volunteers recruited from among students of the University of Warsaw. They had no prior meditation experience but declared an interest in meditation and gave their informed consent for participation in the study. Subjects having previous experience in any type of meditation or having any neurological disorder or medical condition were excluded from the study. The inclusion criteria were consent to participate in the experiment and being a primary language speaker as instructions were delivered in Polish. There were no gender or age restrictions for study participation however subjects were recruited from among the University students resulting in homogeneous age group. The procedures followed were in accordance with the ethical standards of the Helsinki Declaration on human experimentation of 1975, as revised in 1983.

Design and procedure

The experiments were conducted in a bright and quiet room. The subjects sat in comfortable chairs with their eyes closed. The experimental session comprised two conditions separated by rest periods: a meditation podcast (M) – subjects listened to a 15-minute guided mindfulness meditation audiotape, recorded by an experienced mindfulness teacher (Mindfulness-Based Stress Reduction, MBSR); and a passive listening control podcast (P) – subjects listened to a 15 min reading of a piece of literature. The meditation recording was developed based on classic sitting mindfulness practice used in MBSR, which leads participants through a “mindfulness of breathing” exercise. The recording contained instructions such as “notice physical sensations of breathing,” “begin to follow the natural movement of the breath, not trying to change it in any way,” and “stay open and curious about your experience with an accepting attitude.” The control recording was a reading of the novel *Nad Niemnem* (On the Niemen River). Importantly, both recordings used a similar speech speed and number of words. In both recordings a man spoke in a gentle, calm voice; however, the fragment of the novel was devoid of the meditative factor. Besides recording the EEG during the podcast, a fragment of spontaneous EEG was recorded before (2 min) and after (5 min) each part. Subjects were asked to keep their eyes closed and no further specific instruction was given during that time. We consider these short intervals as rest periods (R). Between the recordings, the subjects had a few minutes break, during which they stayed in the seat and had a break from the experiment. The order of conditions during the experiment was the same for all participants., i.e. first M followed by P. The schematic diagram of the experiment is provided in Fig. 1.

After the end of the experiment, subjects responded to a questionnaire about their subjective experiences during the experiment. The questionnaire included the following questions:

1. Did you follow the recording instructions most of the time? (Yes/No)
2. Were you disturbed by the voice on the recording, making it challenging to do the exercise? (Yes/No)
3. Did you feel any discomfort due to the EEG recording? (Yes/No)
4. During the exercise, was your attention recurrently distracted by any other issues not related to the experiment? (Yes/No)
5. Do you remember any special feelings associated with the experiment?

Eleven out of sixteen subjects didn't report any obstacles, while five of them reported some difficulties during the experiment. Subjective feeling of difficulty was indicated by a negative answer to question No. 1, or a positive answer to any of questions 2–4. Question No. 5 was only for information and was not taken into account. We assumed that distraction during meditation can occur easily and is a part of the meditation experience, especially in beginners. Accordingly we analyzed the whole group of subjects together. However, to check whether five subjects who reported difficulties could have changed the outcome of the experiment, we performed additional analysis based only on a group of subjects who didn't report any distractions. We found very little differences with the results of the whole group, hence we report the results obtained for all subjects together.

We analyzed differences between M and P conditions and differences between rest and M or P conditions. We compared activity during the rest period before the task with the activity during the task. We did not look at the rest period after the task as it could have been affected by the task itself.

EEG recording

Nineteen electrodes were placed according to the international 10–20 system, referenced to linked earlobes. The ground electrode was placed on the collarbone. All impedances were kept at a

similar value, below 5 kOhm. The signal was acquired using a Porti7 (TMSI) amplifier, sampled at 512 Hz.

EEG analysis

Preprocessing

The data processing and visualization were done in Python using additional libraries: MNE (Gramfort *et al.*, 2013) (for EEG analysis), Connectivipy (for functional connectivity), and Seaborn (for plotting). The EEG data were re-referenced to the average of the mastoids and then zero-phase filtered using a 4th order Butterworth bandpass filter with cutoffs at 0.5 and 45 Hz. The data were then downsampled to 128 Hz.

Artifact rejection

We applied an artifact rejection method based on outlier detection, i.e. the modified Z-score (Iglewicz and Hoaglin, 1993). The median was calculated for each 2 sec. long window, for each of the EEG channels x_i . The windows overlapped by 1.5 sec. Next, the median of the set of x_i values, $\tilde{x}$, was calculated. Then the $MAD = \text{median}|x_i - \tilde{x}|$ factor was computed. Finally, for every window, the modified Z score was obtained according to the formula:

$$M_i = 0.6745 \frac{x_i - \tilde{x}}{MAD}$$

In the next step, epochs were identified as outliers if they had an M score higher than 3.5 in any of the channels. This threshold was chosen experimentally in order to correctly identify artifacts.

Estimation of band power

The Power Spectral Density was estimated by utilizing Welch's method with a Hamming window, 2 s long, and overlapping by 50%. The relative band power was obtained by normalizing the estimates by the total spectral power and summing the obtained spectral densities for the following ranges: 4–8 Hz for the theta, 8–13 Hz for the alpha, and 30–45 Hz for the gamma band. Further, the relative band powers were aggregated by summing them over the channels within the regions of interest (ROI)s:

- Frontal – Fp1, Fp2, F7, F3, Fz, F4, F8
- Central – T3, C3, Cz, C4, T4
- Parietal-Occipital – T5, P3, Pz, P4, T6, O1, O2

Hemispheric asymmetry

The hemispheric asymmetry was assessed by measuring the band power differences between the right and left brain hemispheres in the three regions using the following pairs of channels: Frontal: F8 - F7, F4 - F3; Central: T4 - T3; and Parietal-Occipital O2 - O1. Thus each asymmetry is defined as a difference (right-left) in EEG power.

Functional connectivity

To quantify the functional connectivity for the different frequency bands, we used a full frequency directed transfer function (ffDTF) defined as (Korzeniewska *et al.*, 2003):

$$\eta_{ij}^2(f) = \frac{|H_{ij}(f)|^2}{\sum_f \sum_{m=1}^k |H_{im}(f)|^2} \quad (1)$$

where $H_{ij}(f)$ is an element of the transfer matrix of the system for frequency f , obtained from a multivariate autoregressive model fitted to the data; and k is the number of channels. The ffDTF describes the causal influence of channel j on channel i at frequency f . Using this definition, we can compute the outflow of information from the given channel j for frequency f as:

$$\varphi_j(f) = \sum_{i=1}^k \eta_{ij}^2(f) \quad (2)$$

Using function (2), we estimated the average outflow from each ROI by summing the results of function (2) over all the channels comprising the given ROI. Next, the values we obtained were integrated separately within the three frequency bands: theta, alpha, and gamma. The evaluation of η_{ij}^2 was performed on artifact-clean, epoched data using the Yule-Walker algorithm, with the covariance matrices averaged over the epochs.

Statistical analysis

To compare the measures calculated for two conditions, we used a pairwise bootstrap procedure. We utilized the within-subject difference for a given measure, averaged across participants, as the test statistic. For comparison of M and P conditions, in each of the tests, our null hypothesis stated that the mean difference between the M and P conditions is equal to zero. According to this hypothesis, we could randomly switch the condition labels, M and P, within a subject. Thus we obtained a distribution of the test statistic under the null hypothesis by random sampling the participants, with replacement, and then relabeling the conditions within subjects, followed by averaging across participants. We compared the value of the test statistic, as observed with the original labeling of conditions, to the sampling distribution we obtained. To estimate the p-value we calculated percentiles, which approximate the probability of the observed experimental value being sampled, from the sampling distribution. Analysis of the differences between rest and M or P conditions was performed in a similar fashion. For each of the three measures nine tests were done (three ROIs, three frequency bands); this posed a multiple comparisons problem. To address this, we used a false discovery rate (FDR), implemented in the MNE package (Benjamini and Hochberg, 1995). All reported p-values have been FDR corrected.

Results

In the descriptions below, the results concerning band power, asymmetry, and outflow measures are grouped according to the EEG oscillations' frequency range.

Theta band

The evolution of theta band power in the frontal ROIs for each of the subjects is presented in Fig. 2. We can observe small time-variability of power during both target and control tasks. Also, theta power is rather similar between the subjects. On the other hand, there is no evident trend of power being higher in one task vs. the other.

The average theta band power distributions are presented in Fig. 3A, and the boxplots of the differences between the conditions are plotted in Fig. 3B. The power differences between the conditions, M-P, were positive in the Frontal and Central ROIs but were not significant for any of the three ROIs ($p > 0.05$). The difference in hemispheric asymmetry between the conditions was not significant either ($p > 0.05$ for all ROIs). The only significant difference in the theta range was decreased activity outflow from the Central (mean = -0.40 , $p = 0.034$) and Parietal-Occipital (mean = -0.25 , $p = 0.049$) ROIs. The distributions of the activity outflow for each ROI in both conditions are shown in Fig. 3C. The boxplots of the outflow differences are shown in Fig. 3D. Details of the test results are presented in Table 1 in the Appendix.

Additionally, we looked at spectral power, asymmetry, and activity flow differences between rest (R) and task periods. For the M task, significant differences were found in the activity outflow from the Frontal (mean = 0.27 , $p = 0.01$) and Central ROIs (mean = 0.42 , $p = 0.02$), which were higher during rest, and in the Parietal-Occipital power asymmetry (mean = 0.004 , $p = 0.047$) corresponding to higher asymmetry during R. For the P task, we observed no significant differences in the theta band.

Alpha band

The evolution of alpha band power in the frontal ROI for each of the subjects is presented in Fig. 4. In a similar way to the theta band, there is no clear pattern regarding the differences between conditions. We can also see that power in the alpha band changes during the course of the experiment; for example, in subjects 5 and 11 the alpha power is higher during the first half of the experiment and decreases in the second half in both tasks. Similar behavior is seen in subjects 3 and 9 but only in the P task.

The average alpha band power distributions are shown in Fig. 5A. The boxplots of the differences between the conditions are plotted in Fig. 5B. The differences between the conditions for alpha band power were negative but were not significant for any of the ROIs ($p > 0.05$). The measures of asymmetry were also no different for any the ROIs ($p > 0.05$). However, we obtained significant differences in the alpha band activity outflow, which was less in the M than in the P condition for the Frontal (mean = -0.15 , $p = 0.049$), Central (mean = -0.51 , $p = 0.02$) and Parietal-Occipital (mean = -0.77 , $p = 0.038$) ROIs. The distribution of the outflow for each ROI for both conditions in the group of subjects is visualized in Fig. 5C. The boxplots of the outflow differences between the conditions are shown in Fig. 5D. Details of the test results are presented in Table 2 in the Appendix.

A comparison between rest and meditation task in the alpha band showed significant differences in power in the Central ROI (mean = 0.024 , $p = 0.042$) and activity outflow in the Frontal (mean = 0.31 , $p = 0.005$) and Central (mean = 0.58 , $p = 0.014$) ROIs, with higher values in R than in M condition. For the P and R task comparison, we obtained significantly higher values of alpha band power during R for all ROIs - Frontal (mean = 0.036 , $p = 0.009$), Central (mean = 0.035 , $p = 0.013$), Parietal-Occipital (mean = 0.056 , $p = 0.013$) and activity outflow in the Central ROI (mean = 0.71 , $p = 0.011$).

Gamma band

An analogous analysis to that of the theta and alpha band was conducted for the gamma band measures, but none of the measures yielded significant differences between conditions M and P. The values of the statistical tests are reported in Table 3 in the Appendix. Similarly, we observed no statistically significant differences between R and M in the gamma frequency band. However, we

obtained significantly higher activity outflow in the Frontal ROI in the R as compared to P condition (mean = 0.34, $p = 0.036$).

Discussion

In this article, we identify and characterize changes in EEG activity that are related to guided mindfulness of breathing meditation practice in the first-time meditators. We compared this activity with the passive listening of a quiet podcast and also looked at differences between rest periods and the tasks. We analyzed neural oscillations in the theta, alpha, and gamma frequency bands, in which changes in spectral power during mindfulness meditation have frequently been observed (see Lee *et al.*, 2018 for meta-analysis). In addition to well-established measures of EEG band power, we also calculated hemispheric band power asymmetry, and the transmission of brain activity between EEG channels, for different frequency bands.

We did not detect any significant differences between the two tasks in EEG power in any of the bands. However, alpha band power was lower during M than during the P task at the margin of statistical significance ($p < 0.07$) in the Central and Parietal-Occipital ROIs (Table 2). It might indicate higher involvement in the M task. Additionally, we noticed that during the M and P tasks alpha power in the Frontal region had a variable time course with a predominant trend of power decrease over time (Fig. 4). It might suggest that the subjects were progressively engaged to both tasks and were not fully resting at that time. It is in agreement with comparison of alpha power during the M and P tasks and the resting period before the task. During both tasks alpha power was significantly lower than during rest.

Lack of differences in EEG oscillations between the M task as compared with the P task might suggest that entering a meditative state during the guided breathing meditation session conducted in the EEG laboratory may impose a serious challenge to first-time meditators. It is consistent with literature findings indicating that meditation-related changes in EEG band power were mainly observed in participants who had long-term meditation experience (Aftanas and Golocheikine, 2001; Pasquini *et al.*, 2015).

Furthermore, we did not find any inter-hemispheric power differences related to the meditation tasks. Henriques and Davidson (1991) proposed a model in which two major motivational systems were postulated: the appetitive and aversive systems. These systems correspond, respectively, to positive and negative effects; lead to approach or withdrawal behavior in response to environmental stimuli, and their balance was claimed to be reflected in the differential activity in resting-state EEG. Indeed, the tendency for withdrawal-related, aversive responses was found to be associated with stronger, relative right-hemispheric anterior activity (e.g. Harmon-Jones *et al.*, 2010; Keune *et al.*, 2011). This pattern of frontal alpha asymmetry was repeatedly found in individuals with a current episode or history of depression (Gotlib *et al.*, 1998; Henriques and Davidson, 1990, 1991). However, the results of two recent meta-analyses have questioned the value of alpha asymmetry as a marker of depression (Kaiser *et al.*, 2018; and van der Vinne *et al.*, 2017). On the other hand, significant increases in left-sided anterior activation were found in a group of meditators following eight weeks of meditation practice compared with the non-meditators (Davidson *et al.*, 2003). However, a recently published study reported a lack of EEG changes in frontal alpha asymmetry in people with major depression disorder following mindfulness training (Szumska *et al.*, 2020).

The only significant difference in EEG signals between conditions was a change in the transmission of activity in the theta and alpha band, as measured by the ffDTF function. Our initial working hypothesis was that the outflow of activity from the Frontal regions should increase during meditation conditions, as increased activation of frontal cortical areas has been observed during

meditation (contrasted to arithmetic) (Hölzel *et al.*, 2007). We found a decrease in the theta activity outflow from the Central and Parietal-Occipital regions and decrease of alpha activity outflow from all ROIs during meditation when compared to passive listening. We think that overall decrease of activity flow in the alpha band might be related to decreased power in this band during meditation exercise. Although power differences between conditions were not statistically significant according to the bootstrap tests for differences in mean values, and the confidence intervals for median values, we did observe a trend in power reduction in the alpha band in these regions during meditation task (cf. negative median values for the difference between M and P in Fig. 5B).

Alpha band activity is commonly associated with relaxed wakefulness (Niedermeyer and Lopes da Silva, 1999). We might hypothesize that in our study novice meditators were more alert and focused on the instructions during the meditation condition than when listening to the podcast. This should not be surprising, as invoking a state of mindfulness during meditation requires the activation of a number of executive functions, both attentional and non-attentional, such as response inhibition and shifting (Holas and Jankowski, 2013). In more expert meditators this activation of executive functions is more automatized and associated with a subjective state of relaxed wakefulness; however, in novices, it might be associated with increased alertness and cognitive effort. Our hypothesis is supported by the observation that alpha power during both tasks was lower than during the resting period, suggesting increased cognitive activity and lack of relaxation during the tasks. Additionally it has been shown by Dissanayaka *et al.* (2015) that a change in attentiveness has a strong impact on the information flow between different brain areas as in our study. On the other hand, their results are not directly comparable as they investigated differences in EEG activities in three distinct states: awake, meditation, and drowsiness.

We should emphasize that although the results of our study related to either EEG power or asymmetry changes during meditation do not confirm the effects described in the literature, there are a number of important differences between the experimental paradigm used in our study and those used by others (reviewed by Tang *et al.*, 2015). Meditation studies often employ cross-sectional design, where a group of long-term meditators is compared with a control group with limited or no meditation experience. In our study, we aimed to investigate whether some changes in EEG signals may be manifested in first-time meditators. Accordingly, we used a longitudinal design, in which data from the same group of subjects is collected over different time points. We compared various measures of EEG activity in naive subjects during a guided meditation and during a carefully selected control task. Our main finding, namely a difference in activity flow in the theta and alpha band shows that the subjects were indeed differentially engaged by the target and control tasks. Decreased alpha band activity outflow, possibly related to reduced power in this band, suggests that a first meditation attempt may be associated with increased alertness and cognitive activity, which may prevent an internalized meditative experience being entered. Accordingly, our result may be relevant for novice meditators and meditation instructors by showing that increased cognitive effort may hinder the ability to achieve the desired state during the initial stages of meditation training.

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Appendix

Table 1. Bootstrap-test results for theta band. P-value calculated using bootstrap pairwise test with FDR correction. Experimental mean is the average value of the difference between M and P in the population. Results are presented for three calculated indicators and ROIs. As a significance level we assumed $\alpha = 0.05$. Significant differences are in bold and marked by an asterisk.

Indicator	Localization	p-value	Experimental mean
Power	Frontal	0.942	0.004
	Central	0.942	0.003
	Parietal-Occipital	0.942	0.008
Asymmetry	Frontal	0.264	-0.002
	Central	0.264	-0.004
	Parietal-Occipital	0.778	0.000
DTF	Frontal	0.075	-0.117
	Central	0.034	-0.396*
	Parietal-Occipital	0.049	-0.245*

Table 2. Bootstrap-test results for alpha band. P-value calculated using bootstrap pairwise test after FDR correction. Experimental mean is the average value of the difference between M and P in the population. Results are presented for three calculated indicators and ROIs. As a significance level we assumed $\alpha = 0.05$. Significant differences are in bold and marked by an asterisk.

Indicator	Localization	p-value	Experimental mean
Power	Frontal	0.129	-0.026
	Central	0.068	-0.032
	Parietal-Occipital	0.058	-0.049
Asymmetry	Frontal	0.422	-0.002
	Central	0.264	-0.008
	Parietal-Occipital	0.264	-0.006
DTF	Frontal	0.049	-0.153*
	Central	0.020	-0.511*
	Parietal-Occipital	0.038	-0.768*

Table 3. Bootstrap-test results for gamma band. P-value calculated using bootstrap pairwise test after FDR correction. Experimental mean is the average value of the difference between M and P in the population. Results are presented for three calculated indicators and ROIs. As a significance level we assumed $\alpha = 0.05$.

Indicator	Localization	p-value	Experimental mean
Power	Frontal	0.909	0.000
	Central	0.935	0.001
	Parietal-Occipital	0.648	-0.002
Asymmetry	Frontal	0.929	0.001
	Central	0.929	0.010
	Parietal-Occipital	0.929	0.005
DTF	Frontal	0.875	0.079
	Central	0.715	0.027
	Parietal-Occipital	0.481	-0.036



Figure 1. The schematic diagram of the experiment. Each experiment consisted of two recording sessions separated by a few minutes break. Each recording began with a 2 min rest period followed by condition M or P and was then concluded by a 5 min rest period.

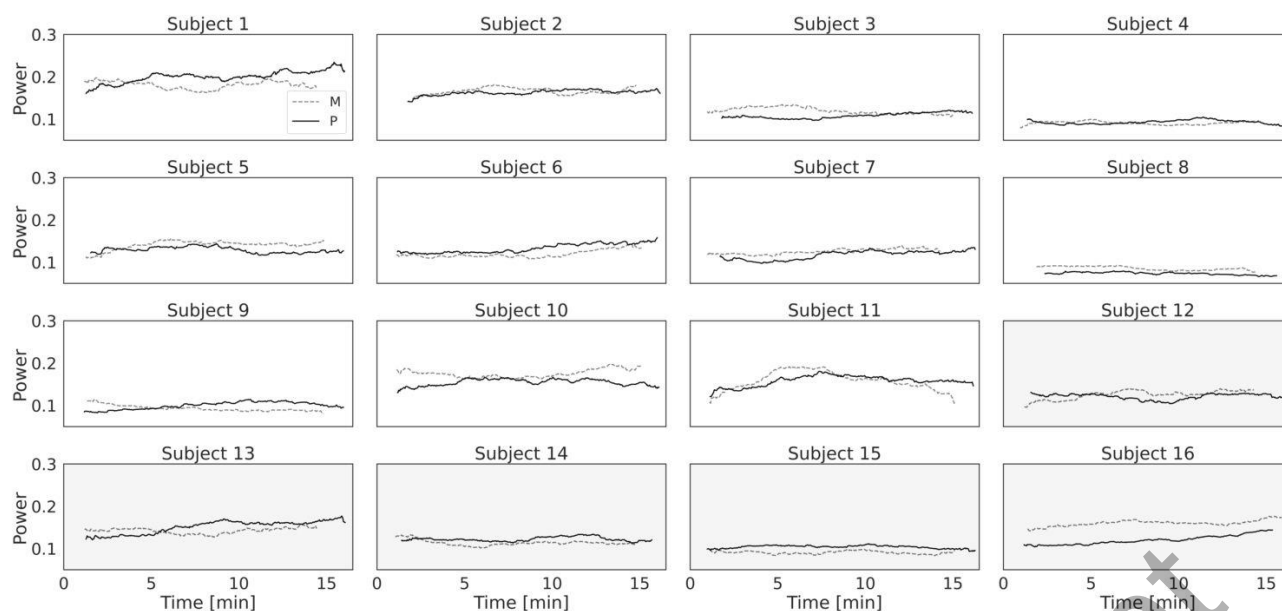


Figure 2. The time course of theta band power from the frontal ROIs for all subjects. Each point corresponds to an average value over 120 s under M (grey dashed line) and P (black solid line) conditions. Shaded windows correspond to subjects (no. 12-16) that reported some difficulties in the questionnaire. The picture shows similar values of theta power during both tasks in all subjects.

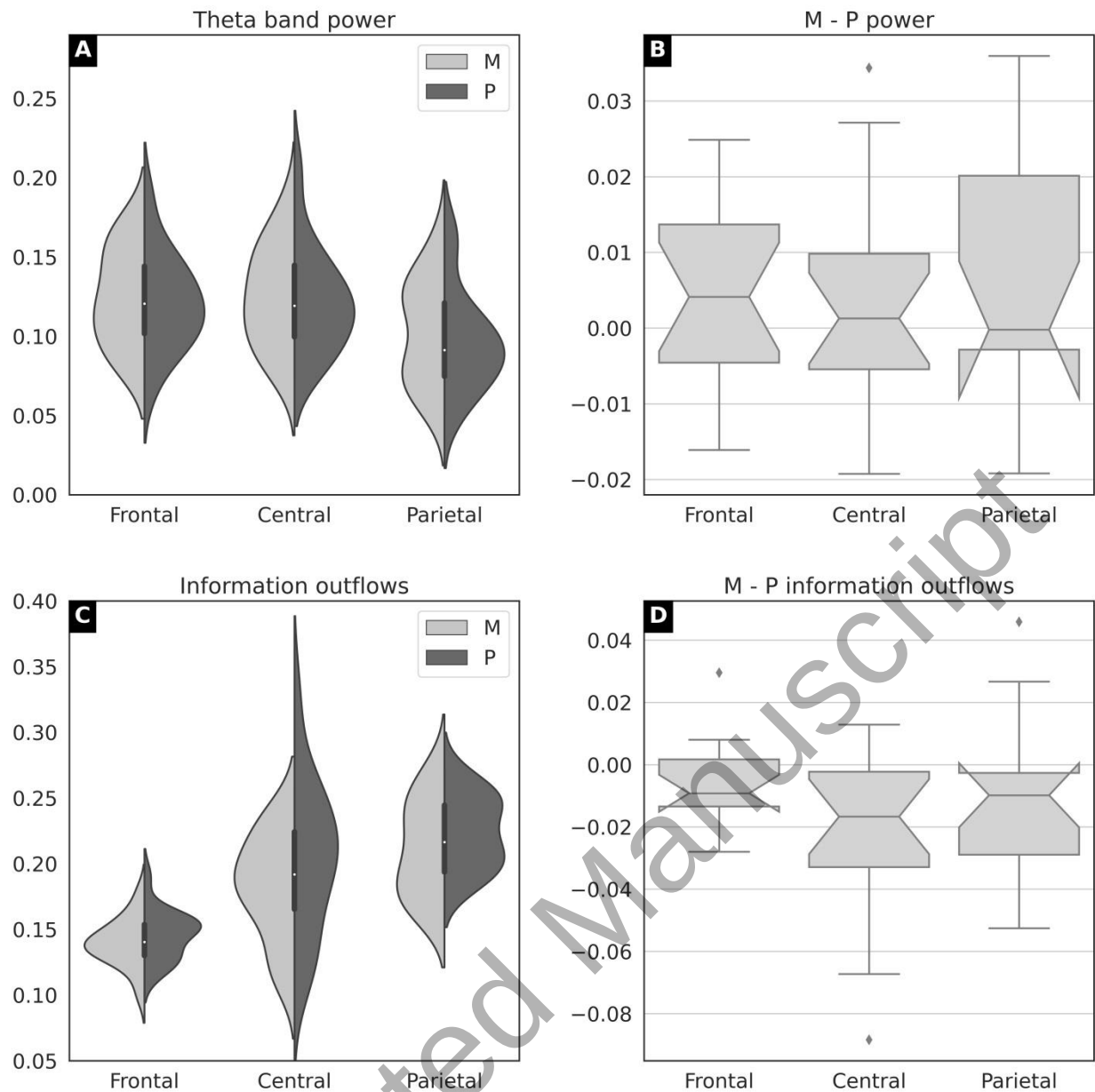


Figure 3. Theta band results for the ROIs; A – distributions of theta band power across subjects (M – light gray, P – dark gray); B – boxplot of pairwise differences (M - P) in the theta band between conditions and across subjects; C – distributions of information outflows across subjects (M – light gray, P – dark gray); D – boxplot of outflows, pairwise differences between M and P. In the plots (A,C), the black solid lines show the interquartile range, while the white dots show medians. The boxplots (B, D) show medians with upper and lower quartiles; notches mark the 95% confidence interval for the median; the whiskers show the extent of the data within the range of 1.5 times the interquartile range, from the first and third quartiles respectively; individual dots denote outliers.

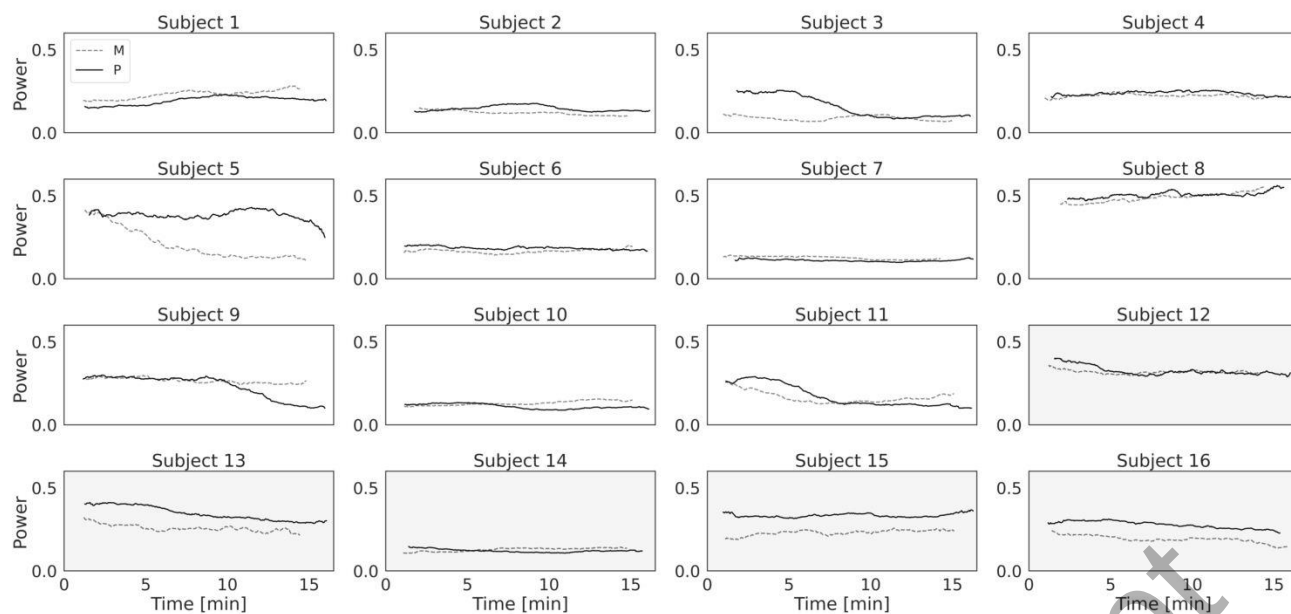


Figure 4. Time course of alpha band power from the frontal ROIs for all subjects. Each point corresponds to an average value over 120 s under M (gray dashed line) and P (black solid line) conditions. Shaded windows correspond to subjects (no. 12-16) that reported some difficulties in the questionnaire. The picture shows some time-variability of alpha power during both tasks in all subjects.

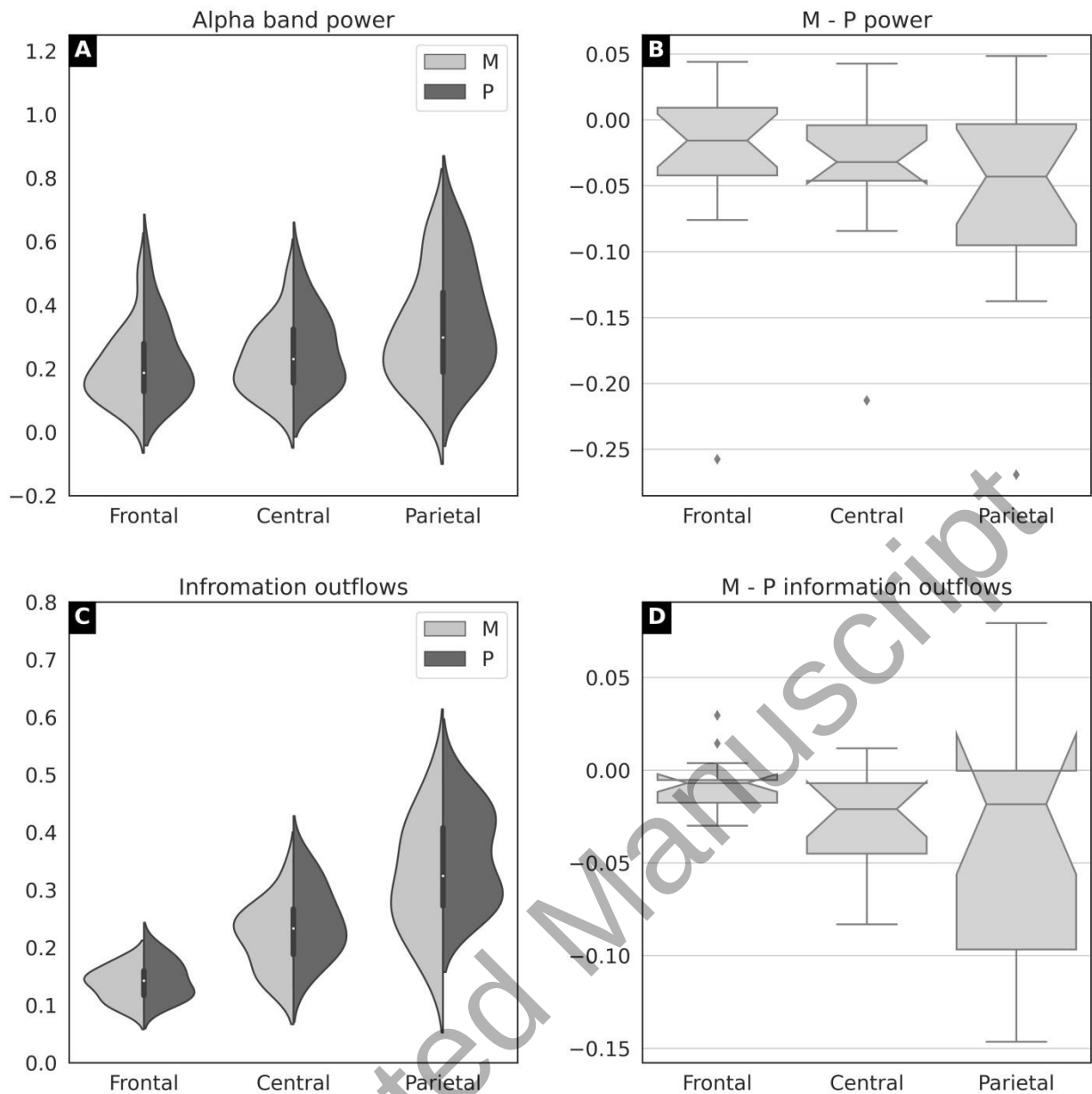


Figure 5. Alpha band results for the ROIs; A – distributions of power across subjects (M – light gray, P – dark gray); B – boxplot of pairwise differences (M - P) in alpha band power between conditions and across subjects; C – distributions of information outflows across subjects (M – light gray, P – dark gray); D – boxplot of outflows, pairwise differences between M and P. In the plots (A, C), the black solid lines show the interquartile range, while the white dots show medians. The boxplots (B, D) show medians with upper and lower quartiles; notches mark the 95% confidence interval for the median; the whiskers show the extent of the data within the range of 1.5 times the interquartile range, from the first and third quartiles respectively; individual dots denote outliers.